CSRI-Product Safety Spotlight

Executive Summary

The Problem We're Solving

The Challenge

Public safety discourse relies on anecdotes rather than systematic analysis of national injury data. NEISS data exists but remains too complex for public interpretation.

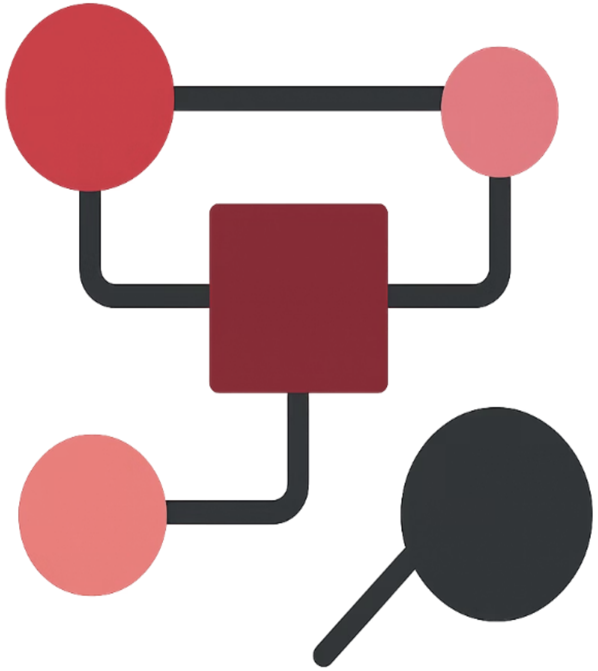
Our Opportunity

Develop predictive models to identify which consumer product injuries result in hospitalization and determine the factors that predict severe medical outcomes.

Primary Objective

Create evidence-based findings for the 2025 Product Safety Spotlight report that inform consumers, manufacturers, and policymakers about genuine safety priorities.

Approach & Methodology



01

Framework

CRISP-DM

Six-Phases

Iterative & Cyclical Process

02

Dataset

NEISS Dataset

7 Million Records

1 GB of Data

03

Challenges

Data Volume

High Class Imbalance (9:1)

High Cardinality Features

Harmonizing Taxonomy Changes

Stratified Split (Year+Target)

04

Evaluation

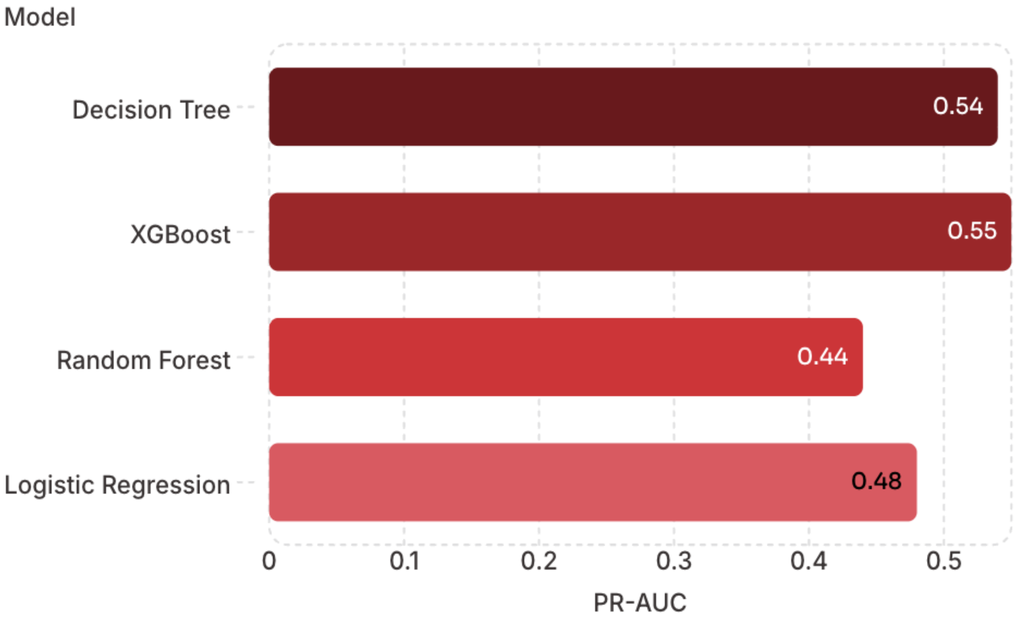
PR-AUC Score

MCC Score

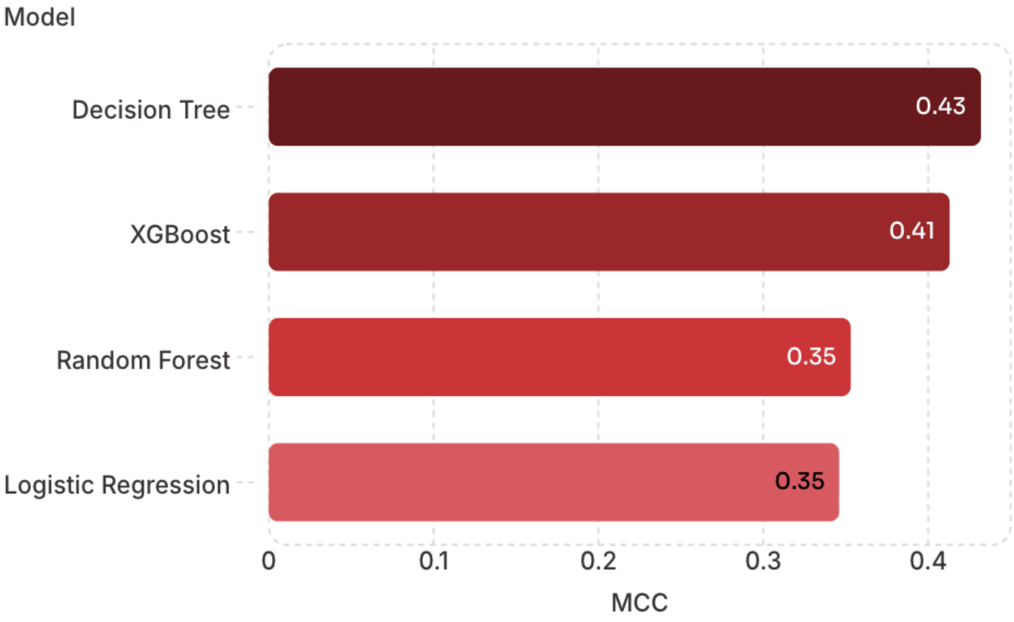
Model Performance Summary

Comparison of PR-AUC and MCC across four candidate models on imbalanced data.

PR-AUC Score



MCC Score



Why Decision Tree was selected

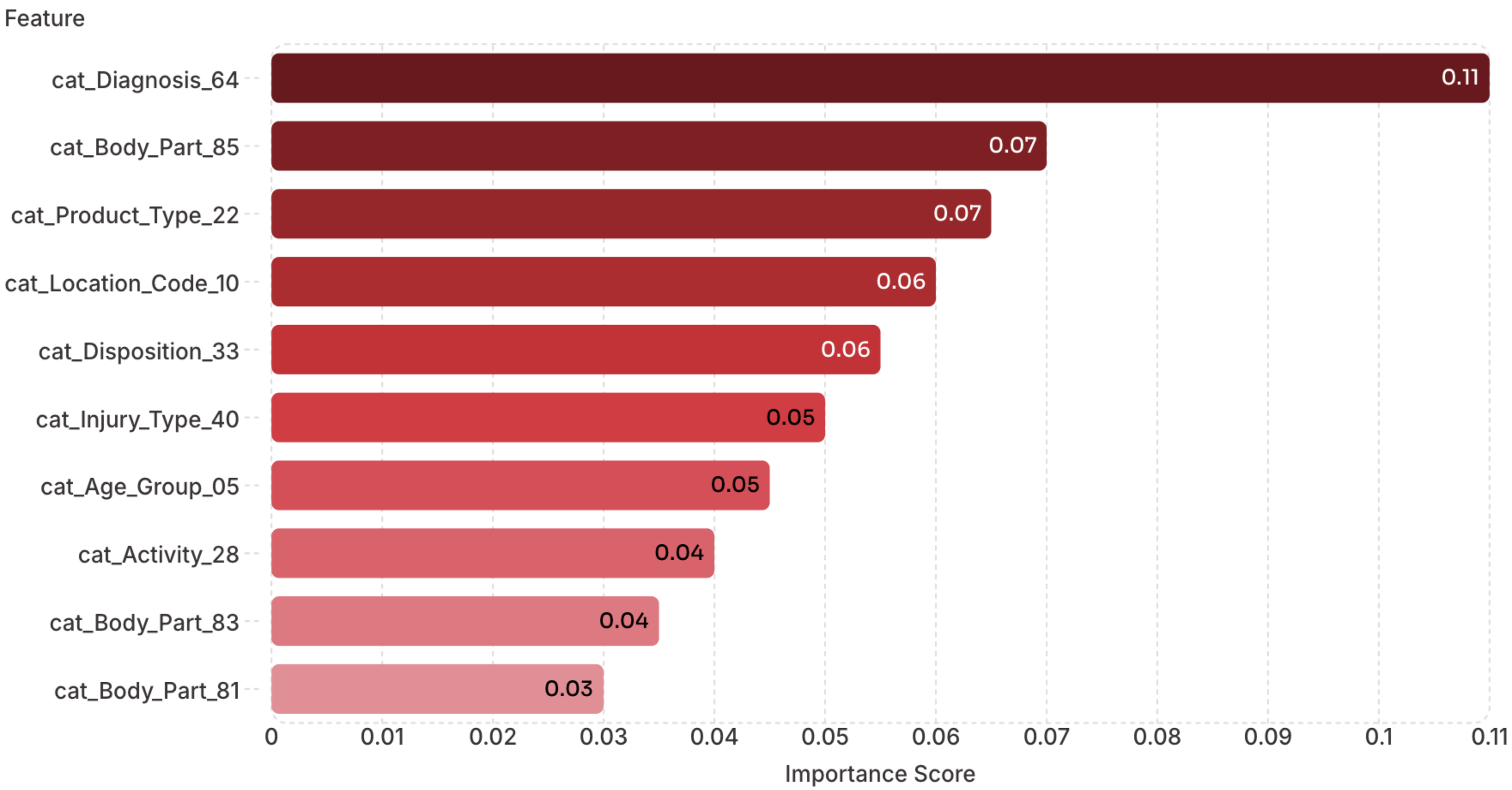
- Best MCC score at 0.432
- Balanced classification at the 0.50 threshold

Why XGBoost was retained

- Best PR-AUC at 0.55
- Useful for ranking high-risk cases

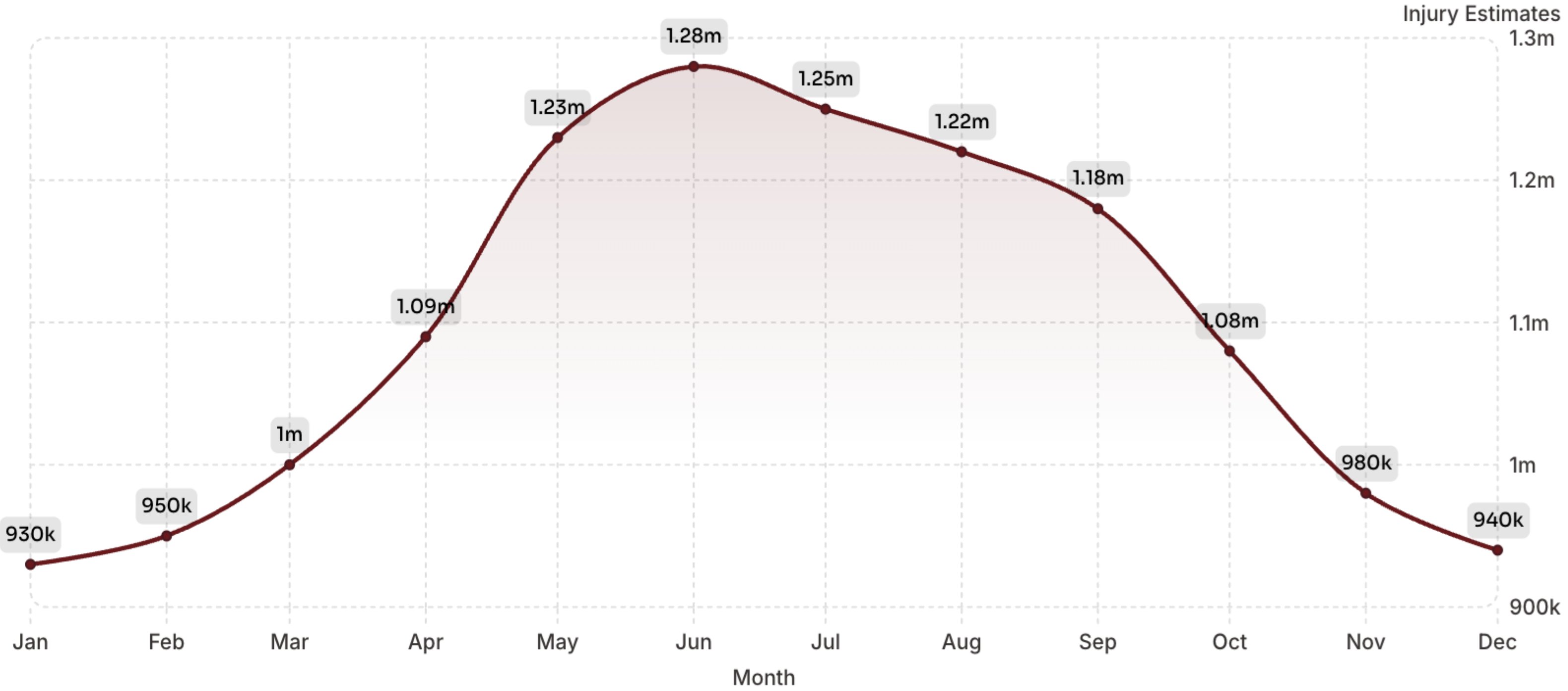
Key insight: Decision Tree was chosen as the primary model because it achieved the best MCC score and provides balanced classification at the 0.50 operational threshold, while XGBoost was designated as the secondary model for its superior PR-AUC performance in ranking high-risk cases.

Model Feature Importance



The XGBoost model identified diagnosis type and body part affected as the strongest predictors of hospitalization. These features should guide clinical assessment and intervention strategies.

Key Finding: Seasonality Patterns



Peak Season

Injury rates peak during **May–September**, with **June** showing the highest severity and an estimated **1.28 million** injuries - Months with high physical and recreational activity.

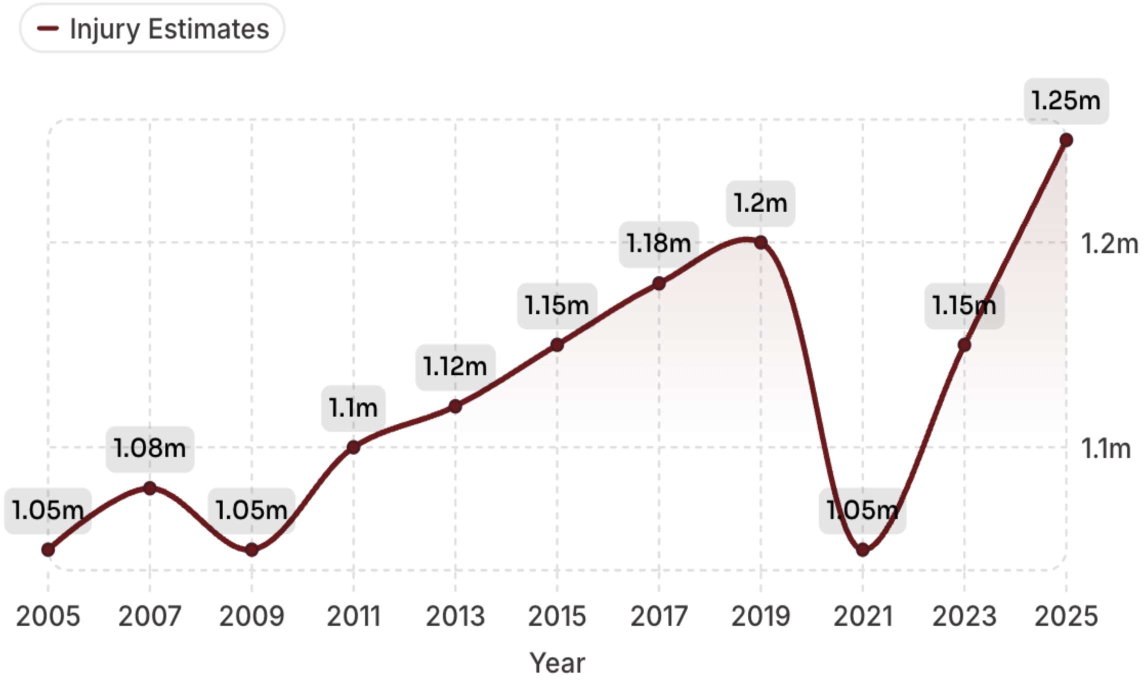
Lowest Risk

Winter months **January–February** show the lowest injury rates, with estimates of **930K** and **950K**, respectively .

Seasonal Swing

Summer months see up to **38% more injuries** than winter lows, reinforcing a strong and consistent seasonality pattern.

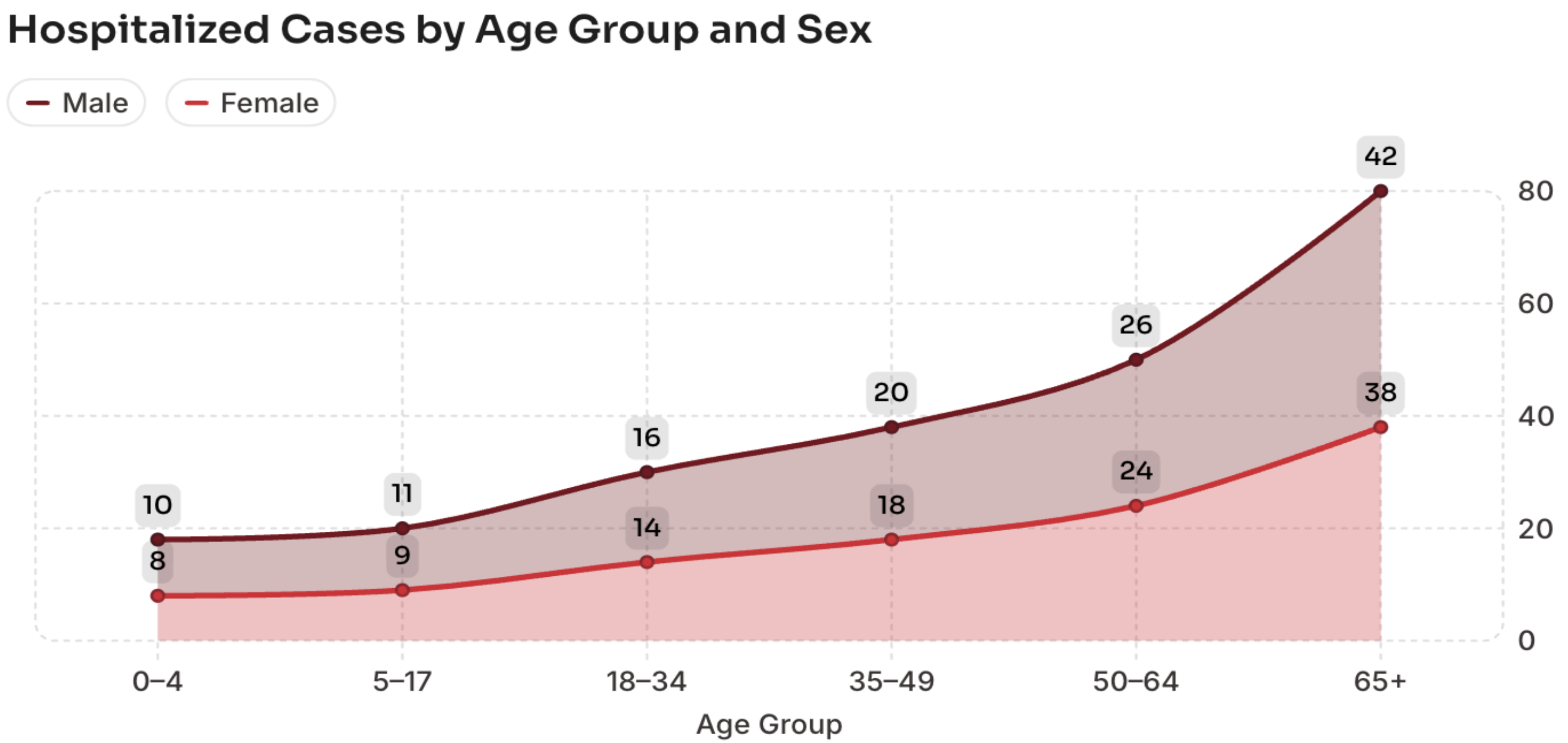
National Injury Trends (2005–2025)



What the Data Shows

National injury estimates remained relatively stable from **2005 to 2019**, peaking at **1.20 million**. A sharp dip occurred in **2020–2021** (down to 1.05 million), followed by a steady recovery projected to reach **1.25 million** by 2025.

Demographic Disparities



Seniors (65+)

28.3% of all hospitalized cases; highest-risk group.

Older Females

Higher hospitalization rates than males in older age groups.

Young Children (0–4)

Lower severity despite higher injury frequency.

High-Risk Product Categories

High Priority

Blankets, Toilets, Rugs, Ladders — high injury volume and high hospitalization severity. Primary targets for intervention.

Hidden Dangers

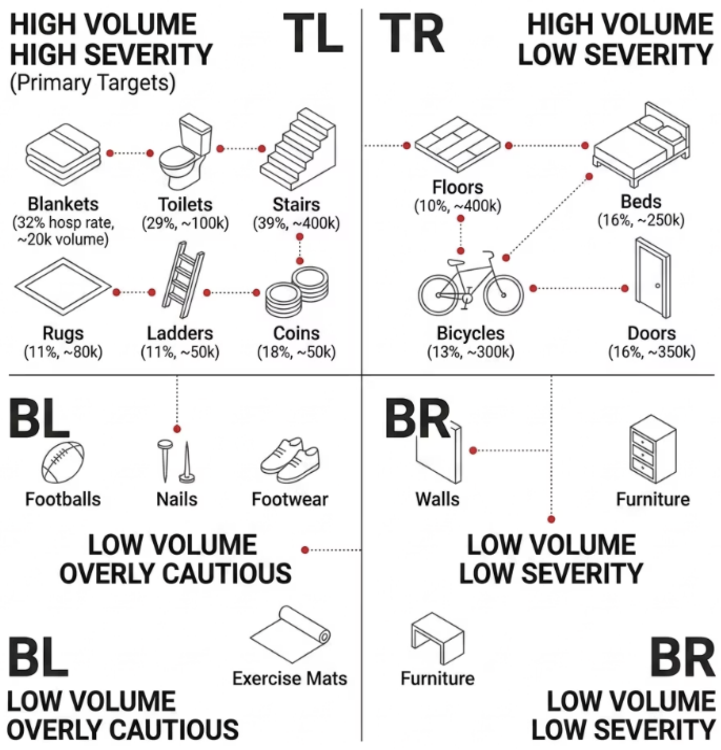
Coins, Porches, Windows — lower volume but high severity. Often overlooked by consumers and regulators.

High Volume

Floors and Stairs — high injury volume with moderate-to-high severity. Stairs show a 39% hospitalization rate.

RISK PRIORITIZATION

Product Risk Matrix: Injury Volume vs. Hospitalization Severity



Products in the **top-left quadrant** combine high injury volume with high hospitalization rates — these are the **primary targets** for safety intervention.

Blankets, Toilets, and Stairs represent the highest-risk combination. Hidden dangers like **coins** have lower volume but alarming severity.

⚠ Floors have the highest injury volume (~400K) but lower severity (10%), making them a volume-driven priority.

Strategic Recommendations

Target High-Risk Groups

Focus safety messaging on **seniors (65+)** and high-risk product categories.

Seasonal Campaigns

Implement safety campaigns with a **May–September focus** to coincide with peak injury months.

Product Design Improvements

Prioritize the redesign of **high-severity items** — Blankets, Toilets, Rugs, Ladders, and Stairs.

Age-Specific Guidance

Develop tailored safety guidance for **vulnerable populations**, especially seniors and young children.

THANK YOU

Thank You

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Questions, feedback, or follow-up are always welcome.